Programing Fundamental Lab#4

Task#1

```c
#include<stdio.h>
int main(){
	float number;
	printf("Enter Number: ");
	if(scanf("%f",&number) !=1){
	printf("You entered invalid Number");
	return 0;}
	if(number<0){
	printf("You entered a negatvie Number");
	}else if(number>0){
	printf("You entered a Positve Number");
	}else{
	printf("You entered a Zero");
	}
return 0;
}
```

Results

Select E:\PF lab\Lab-4\task#1.exe

```
Enter Number: 0
You entered a Zero
--------------------------------
Process exited after 1.556 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#1.exe

```
Enter Number: 5
You entered a Positve Number
--------------------------------
Process exited after 1.62 seconds with return value 0
Press any key to continue . . .
```

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E:\PF lab\Lab-4\task#1.exe

```
Enter Number: -6
You entered a negatvie Number
--------------------------------
Process exited after 1.607 seconds with return value 0
Press any key to continue . . .
```

Task#2

task#1.c [*] task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c task#9.c task#10.c

```c
#include<stdio.h>
int main(){
    int NumberOne,NumberTwo;
    printf("-------------------X------------------\n");
    printf("           Simple Calculator\n");
    printf("-------------------X------------------\n");
    printf("Enter a Number: ");
    scanf("%d",&NumberOne);
    printf("Enter Another Number: ");
    scanf("%d",&NumberTwo);
    int number;
    printf("For Addition enter 1\nFor Subtraction enter 2 \nFor Multiplication enter 3 \nFor division enter 4\n");
    printf("Enter Number of Operation: ");
    scanf("%d",&number);
    switch (number)
    {
    case 1:
        printf("The Sum of Number One and Number two= %d \n",NumberOne+NumberTwo);
        break;
    case 2:
        printf("The Difference of Number One and Number two= %d \n",NumberOne-NumberTwo);
        break;
    case 3:
        printf("The Product of Number One and Number two= %d \n",NumberOne*NumberTwo);
        break;
    case 4:
    if (NumberTwo == 0) {
        printf("Division by Zero is not possible\n");
    } else {
        printf("The Division of Number One and Number two= %d \n", NumberOne / NumberTwo);
    }
    break;

    default:
        printf("You entered invalid Number");

        break;
    }
    return 0;
}
```

Results

```
Enter a Number: 5
Enter Another Number: 5

Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 4
The Division of Number One by Number Two is: 1

--------------------------------
Process exited after 4.755 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#2.exe

```
Enter a Number: 5
Enter Another Number: 5

Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 1
The Sum of Number One and Number Two is: 10

--------------------------------
Process exited after 3.667 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#2.exe

```
Enter a Number: 7
Enter Another Number: 4

Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 2
The Difference between Number One and Number Two is: 3

--------------------------------
```

E:\PF lab\Lab-4\task#2.exe

```
Enter a Number: 3
Enter Another Number: 3

Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 3
The Product of Number One and Number Two is: 9

--------------------------------
```

Programing Fundamental Lab#4

Task#3

task#1.c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c task#8.c task#9.c task#10.c [*] task2.c

```
#include<stdio.h>
int main (){
	int WaterLevel;
	printf("Enter the water level:");
	scanf("%d",&WaterLevel);
	if(WaterLevel>6){
		printf("Evacuation Required");
	return 0;
	}else if(WaterLevel<6 && WaterLevel>4){
		printf("Medium Relief Package");
		return 0;
	}else if(WaterLevel<4 && WaterLevel>2){
		printf("Small Relief Package");
		return 0;
	}else{
		printf("No Relief Required!");
	}
	return 0;
}
```

Results

E:\PF lab\Lab-4\task#3.exe

```
Enter the water level:5
Medium Relief Package
--------------------------------
Process exited after 2.056 seconds with return value 0
Press any key to continue . . .
```

Select E:\PF lab\Lab-4\task#3.exe

```
Enter the water level:3
Small Relief Package
--------------------------------
Process exited after 2.642 seconds with return value 0
Press any key to continue . . .
```

```
Enter the water level:7
Evacuation Required
--------------------------------
Process exited after 1.696 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#3.exe

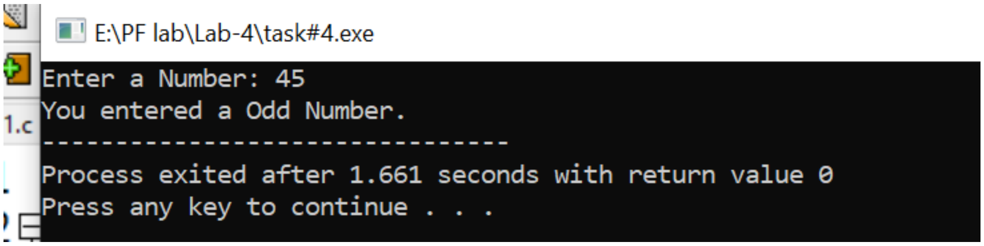
```
Enter the water level:1
No Relief Required!
--------------------------------
Process exited after 2.11 seconds with return value 0
Press any key to continue . . .
```

Task#4

task#1.c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c task#8.c task#9.c task#10.c [*] task2.c

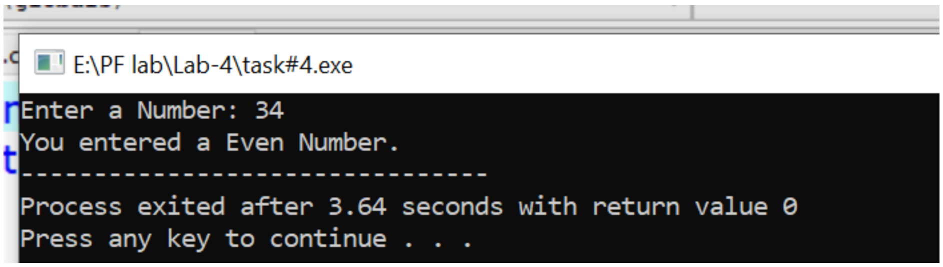
```
#include<stdio.h>
int main(){
	int number;
	printf("Enter a Number: ");
	if(scanf("%d",&number)!=1){
		printf("You enter an invalid Number");
		return 1;
	}
	if(number%2==0){
		printf("You entered a Even Number.");
	}else{
	printf("You entered a Odd Number.");
	}
	return 0;

}
```

Results



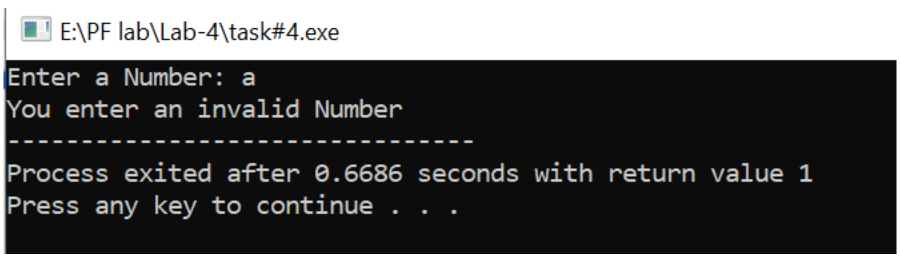
E:\PF lab\Lab-4\task#4.exe

```
Enter a Number: 45
You entered a Odd Number.
--------------------------------
Process exited after 1.661 seconds with return value 0
Press any key to continue . . .
```



E:\PF lab\Lab-4\task#4.exe

```
Enter a Number: 34
You entered a Even Number.
--------------------------------
Process exited after 3.64 seconds with return value 0
Press any key to continue . . .
```



E:\PF lab\Lab-4\task#4.exe

```
Enter a Number: a
You enter an invalid Number
--------------------------------
Process exited after 0.6686 seconds with return value 1
Press any key to continue . . .
```

Task#5

```
#  C:\Program Files (x86)\Dev-Cpp\MinGW64\x86_64-w64-mingw32\include\stdio.h - Ctrl+Click for more info
int main(){
	float Marks;
	printf("Enter your Programming fundamental lab marks: ");
	scanf("%f",&Marks);
	if(Marks>=85 && Marks<=100){
		printf("Your grade is A");
	
	}else if(Marks<=84 && Marks>=75){
		printf("Your grade is B");}
	else if(Marks<=69 && Marks>=55  ){
		printf("Your grade is c");
	}
	else if(Marks<=54 && Marks>=40){
		printf("Your grade is D");
	}else if(Marks<40&& Marks>=0){
	printf("Your grade is f");
	}else{
		printf("Invalid Marks");
	}
	return 0;
}
```

Result

```
E:\PF lab\Lab-4\task#5.exe
Enter your Programming fundamental lab marks: 97
Your grade is A
--------------------------------
Process exited after 1.713 seconds with return value 0
Press any key to continue . . .
```

```
E:\PF lab\Lab-4\task#5.exe
Enter your Programming fundamental lab marks: 76
Your grade is B
--------------------------------
Process exited after 5.704 seconds with return value 0
Press any key to continue . . .
```

```
Enter your Programming fundamental lab marks: 66
Your grade is c
--------------------------------
Process exited after 2.93 seconds with return value 0
Press any key to continue
```

E:\PF lab\Lab-4\task#5.exe

```
Enter your Programming fundamental lab marks: 54
Your grade is D
--------------------------------
Process exited after 4.045 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#5.exe

```
Enter your Programming fundamental lab marks: 39
Your grade is f
--------------------------------
Process exited after 3.17 seconds with return value 0
Press any key to continue . . .
```

Task#6

task#1.c task#2.c task#3.c task#4.c task#5.c [*] task#6.c task#7.c task#8.c task#9.c task#10.c [*] task2.c

```c
#include<stdio.h>
int main(){
	int a, b, c;
	printf("Enter the Value of a: ");
	if(scanf("%d", &a) != 1){
		printf("Invalid Value of A\n");
		return 0;
	}
	printf("Enter the Value of b: ");
	if(scanf("%d", &b) != 1){
		printf("Invalid Value of B\n");
		return 0;
	}
	printf("Enter the Value of c: ");
	if(scanf("%d", &c) != 1){
		printf("Invalid Value of C\n");
		return 0;
	}
	if(a > b && a > c){
		printf("The value of a is largest.\n");
	} else if(b > a && b > c){
		printf("The value of b is largest.\n");
	} else if(c > a && c > b){
		printf("The value of c is largest.\n");
	} else if(a == b && a > c){
		printf("The value of a and b are equal and largest.\n");
	} else if(b == c && b > a){
		printf("The value of b and c are equal and largest.\n");
	} else if(a == c && a > b){
		printf("The value of a and c are equal and largest.\n");
	} else {
		printf("All numbers have same value.\n");
	}
	return 0;
}
```

Results

E:\PF lab\Lab-4\task#6.exe

```
Enter the Value of a: 3
Enter the Value of b: 4
Enter the Value of c: 6
The value of c is largest.

--------------------------------
Process exited after 6.399 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#6.exe

```
Enter the Value of a: 65
Enter the Value of b: 5
Enter the Value of c: 4
The value of a is largest.

--------------------------------
Process exited after 4.286 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#6.exe

```
Enter the Value of a: 6
Enter the Value of b: 90
Enter the Value of c: 7
The value of b is largest.

--------------------------------
Process exited after 5.725 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#6.exe

```
Enter the Value of a: 6
Enter the Value of b: 6
Enter the Value of c: 6
All numbers have same value.

--------------------------------
Process exited after 2.592 seconds with return value 0
Press any key to continue . . .
```

Programing Fundamental Lab#4

Task#7

```
#include<stdio.h>
int main(){
	int ElectricUnit;
	printf("Enter your electricity Unit: ");
	scanf("%d",&ElectricUnit);
	if(ElectricUnit<=100){
		printf("Your electricity bill is: %d",ElectricUnit*10);
	}else if(ElectricUnit>101 &&ElectricUnit<=300){
		printf("Your electricity bill is: %d",ElectricUnit*15);
	}else if(ElectricUnit>301 && ElectricUnit<=500){
		printf("Your electricity bill is: %d",ElectricUnit*20);
	}else if(ElectricUnit>=500){
		printf("Your electricity bill is: %d",ElectricUnit*25);
	}
	return 0;
}
```

Results

E:\PF lab\Lab-4\task#7.exe

```
Enter your electricity Unit: 56
Your electricity bill is: 560
--------------------------------
Process exited after 1.307 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#7.exe

```
Enter your electricity Unit: 150
Your electricity bill is: 2250
--------------------------------
Process exited after 1.817 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#7.exe

```
Enter your electricity Unit: 450
Your electricity bill is: 9000
--------------------------------
Process exited after 2.649 seconds with return value 0
Press any key to continue . . .
```

```
ude<stdio.h>
```

E:\PF lab\Lab-4\task#7.exe

```
Enter your electricity Unit: 600
Your electricity bill is: 15000
--------------------------------
Process exited after 1.094 seconds with return value 0
Press any key to continue . . .
```

Task#8

```
.c  task#2.c  task#3.c  task#4.c  task#5.c  task#6.c  task#7.c  task#8.c  task#9.c  task#10.c  [*] task2.c
#include <stdio.h>
#include <math.h>

int main() {
    int NumberOne, NumberTwo, operation;

    printf("Enter a Number: ");
    scanf("%d", &NumberOne);

    printf("Enter Another Number: ");
    scanf("%d", &NumberTwo);

    printf("\nChoose your operation:\n");
    printf("For Addition press 1\n");
    printf("For Subtraction press 2\n");
    printf("For Multiplication press 3\n");
    printf("For Division press 4\n");
    printf("For Modulus press 5\n");
    printf("For Power (NumberOne raised to NumberTwo) press 6\n");
    printf("Enter Number of Operation: ");
    scanf("%d", &operation);

    switch (operation) {
        case 1:
            printf("The Sum of Number One and Number Two is: %d\n", NumberOne + NumberTwo);
            break;
        case 2:
            printf("The Difference between Number One and Number Two is: %d\n", NumberOne - NumberTwo);
            break;
        case 3:
            printf("The Product of Number One and Number Two is: %d\n", NumberOne * NumberTwo);
            break;

            break;
        case 4:
            if (NumberTwo == 0) {
                printf("Division by Zero is not possible.\n");
            } else {
                printf("The Division of Number One by Number Two is: %d\n", NumberOne / NumberTwo);
            }
            break;
        case 5:
            if (NumberTwo == 0) {
                printf("Modulus by Zero is not allowed.\n");
            } else {
                printf("The Modulus of Number One and Number Two is: %d\n", NumberOne % NumberTwo);
            }
            break;
        case 6:
            printf("Number One raised to the power of Number Two is: %.2f\n", pow(NumberOne, NumberTwo));
            break;
        default:
            printf("You entered an invalid operation number.\n");
    }

    return 0;
}
```

Programing Fundamental Lab#4

Task#9

```
#include<stdio.h>;
int main(){
    int RainFall=1445,RiverFlow=5450;

    if(RainFall<50 && RiverFlow<200){
        printf("Low Risk");
        return 0;
    }else if(RainFall>50 && RainFall<100 && RiverFlow>200 && RiverFlow<500){
        printf("Moderate Risk");
        return 0;
    }else if(RainFall>100 && RainFall<150 && RiverFlow>500 && RiverFlow<800){
        printf("High Risk");
        return 0;
    }else if(RainFall>150 && RiverFlow>800 ){
        printf("Severe Risk  -Evacuate!");
    }
    return 0;
}
```

Task#10

```
#include<stdio.h>
#include<stdlib.h>
int main(){
    int AkhlaqManners,HonestyTrustworthiness,PrayerRegularity,
    Fasting,ZakatCharity,SocialBehavior,ConflictSkills;

    printf("-----------------x--------------------x------------\n");
    printf("We are Calculating the Character of a man as per Islam\n");
    printf("-----------------x--------------------x------------\n");

    printf("Enter Akhlaq & Manners range 0-10:  ");
    scanf("%d",&AkhlaqManners);
    // system(cls);
    printf("Enter Honesty &Trustworthiness range 0-10:  ");
    scanf("%d",&HonestyTrustworthiness);
    printf("Enter Prayer Regularity range \n0 for Irregular,\n 1 for Regular: ");
    scanf("%d",&PrayerRegularity);
    printf("Enter Fasting range \n0 for Never,\n1 for Somtimes\n2 for Always : ");
    scanf("%d",&Fasting);
    printf("Enter Zakat & Charity range 0-10:  ");
    scanf("%d",&ZakatCharity);
    printf("Enter Social Behavior range 0-10:  ");
    scanf("%d",&SocialBehavior);
    printf("Enter Conflict Skills range 0-10:  ");
    scanf("%d",&ConflictSkills);
    if (PrayerRegularity == 0) {
    PrayerRegularity = 0;
    } else {
    PrayerRegularity = 10;
    }
```

```c
if (Fasting == 0) {
Fasting = 0;
} else if (Fasting == 1) {
Fasting = 5;
} else {
Fasting =10;
}
    system("cls");
    printf("------------------x-----------------------x-------------\n");
    printf("Character of a man as per Islam\n");
    printf("------------------x-----------------------x-------------\n");
float CharacterIndex= (AkhlaqManners * 2.5)+(HonestyTrustworthiness*2.0)+(PrayerRegularity*1.0)+(Fasting*1.5)+(ZakatCharity*1.0)+(Soci

if(CharacterIndex>=85&&CharacterIndex<=100){
    printf("Character: Excellent Muslim Character. \n");
    printf("Remarks: Role Mode for society");
    return 0;
}else if(CharacterIndex>=70&&CharacterIndex<85){
    printf("Character: Good Muslim Character. \n");
    printf("Remarks: Practicing Believer.");
    return 0;
}else if(CharacterIndex>=50&&CharacterIndex<70){
    printf("Character: Average Muslim Character. \n");
    printf("Remarks: Needs minor improvement.");
    return 0;
}else if(CharacterIndex>=30&&CharacterIndex<50){
    printf("Character: Needs Improvement. \n");
    printf("Remarks: Work on Akhlaq and Ibadad.");
    return 0;
}else if(CharacterIndex<30){
    printf("Character: Weak Character. \n");
    printf("Remarks: Require serious guidance");
    return 0;
}else{
    printf("Invalid Input");
}
return 0;
```

Github link

https://github.com/vishnani/Lab-4-task